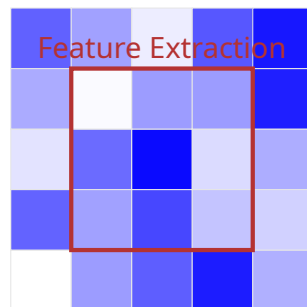


The Mathematics of Vision

*A Comprehensive Module on Image Processing, Analysis, and
Kernel-Based Classification*

Prepared for the Student of Algorithms



January 23, 2026

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Chapter 1

Foundations of Digital Representation

1.1 The Discrete Nature of Digital Images

To understand computer vision, one must first strip away the semantic meaning of an image. To a human, a photograph might portray a sunset or a familiar face. To a computer, an image is nothing more than a discrete function of light intensity defined over a two-dimensional domain.

Let $f(x, y)$ be a continuous function representing the light intensity at spatial coordinates (x, y) . However, computer memory is finite, necessitating that we sample this continuous signal into a discrete grid. This process, known as digitization, yields a matrix I of dimensions $M \times N$.

$$I = \begin{pmatrix} I(0, 0) & I(0, 1) & \cdots & I(0, N-1) \\ I(1, 0) & I(1, 1) & \cdots & I(1, N-1) \\ \vdots & \vdots & \ddots & \vdots \\ I(M-1, 0) & I(M-1, 1) & \cdots & I(M-1, N-1) \end{pmatrix}$$

1.1.1 Visualizing the Pixel Grid

The concept of a "pixel" is abstract. Mathematically, it is a scalar value located at an integer coordinate. Below is a visualization of a simple 5×5 image patch. The number represents the intensity (0-255), and the background color represents how it appears to the human eye.

The Matrix Representation

25	50	200	240	50
40	60	210	250	60
30	55	205	230	55
10	20	40	50	30
0	10	20	30	10

Notice: High values (white) vs Low values (black)

Figure 1.1: An image is fundamentally a matrix of integers.

1.2 Color Spaces: Vectors at Every Point

While a grayscale image is a scalar field, a color image is a vector field. In the RGB color model, every pixel is a vector $\mathbf{v} \in \mathbb{R}^3$.

$$I(x, y) = \begin{bmatrix} R(x, y) \\ G(x, y) \\ B(x, y) \end{bmatrix}$$

We can visualize this not as a single matrix, but as three stacked planes (channels).

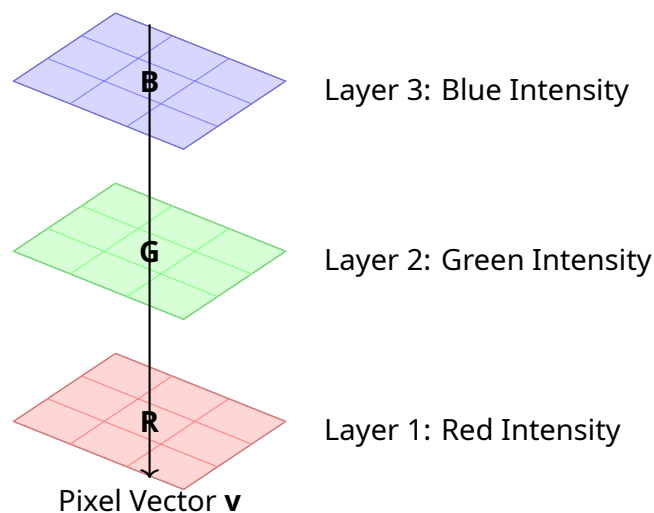


Figure 1.2: Decomposition of a Color Image into RGB Channels.

1.3 From Matrix to Feature Vector: Flattening

For many machine learning algorithms (such as Multilayer Perceptrons or Classical Regression), we cannot feed the 2D (or 3D) image structure directly. We must transform the spatial grid into a linear Feature Vector.

Consider an image of Height R (rows) and Width C (columns). The total number of pixels is $N = R \times C$. Each pixel P_i (where $0 \leq i < N$) is a vector containing color information. For RGB, $\mathbf{x}_i \in \mathbb{R}^3$:

$$\mathbf{x}_i = \begin{bmatrix} r_i \\ g_i \\ b_i \end{bmatrix}$$

where r_i, g_i, b_i are the intensity values for Red, Green, and Blue channels respectively.

The process of **Flattening** involves reading the pixels in a specific order (typically raster scan: row by row, left to right) and concatenating their vector components. The final data point for a supervised learning task includes the feature vector followed by the class label y .

Final Vector Dimension:

$$D = (R \times C \times 3) + 1$$

The structure is: $[r_0, g_0, b_0, r_1, g_1, b_1, \dots, r_{N-1}, g_{N-1}, b_{N-1}, y]$.

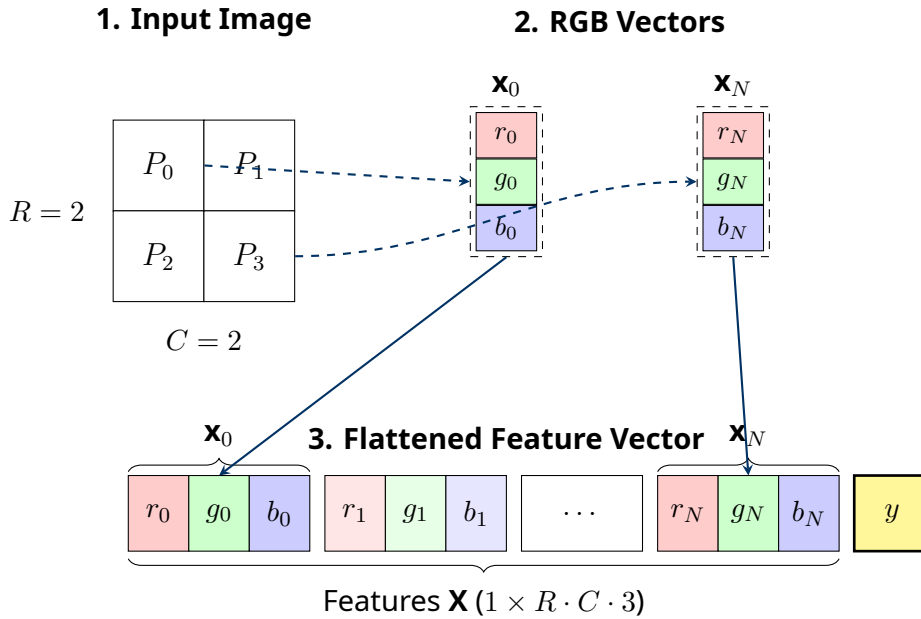


Figure 1.3: Step-by-step transformation from Image Grid to Feature Vector.

Chapter 2

Statistical Analysis and Preprocessing

Statistical analysis provides the mathematical framework for understanding the global properties of an image. By treating pixel intensities as random variables, we can apply probability theory to analyze brightness, contrast, and information content.

2.1 Histogram Analysis: The Probability of Brightness

A histogram is an approximation of the Probability Mass Function (PMF) of the image's intensity. Let r_k be a discrete intensity level (e.g., $k \in [0, 255]$) and n_k be the number of pixels with intensity r_k . The unnormalized histogram is the function $h(r_k) = n_k$. To treat this as a probability distribution, we normalize by the total number of pixels MN :

$$p(r_k) = \frac{n_k}{MN}$$

Here, $p(r_k)$ represents the probability that a randomly chosen pixel has intensity r_k . Note that $\sum_{k=0}^{L-1} p(r_k) = 1$.

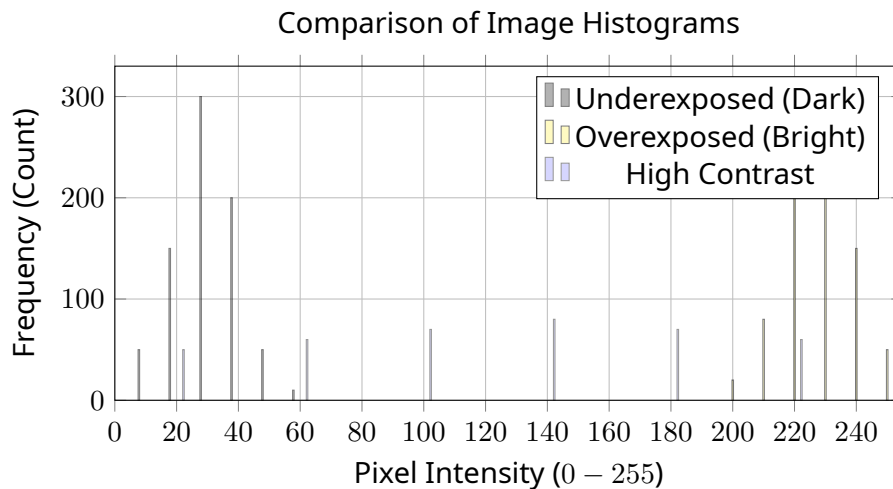


Figure 2.1: Histogram shapes reveal exposure and contrast properties.

2.2 Central Tendency: The Mathematics of Brightness

The "brightness" of an image is best described by measures of central tendency. The most common measure is the **Mean (Average) Intensity**, denoted as μ or $E[r]$.

$$\mu = \sum_{k=0}^{L-1} r_k p(r_k)$$

If $\mu < 128$, the image is generally dark (low-key). If $\mu > 128$, the image is bright (high-key). A secondary measure is the **Median**, which is less sensitive to outliers (such as salt-and-pepper noise). The median m is the value such that $P(r \leq m) \geq 0.5$.

2.3 Statistical Dispersion: The Mathematics of Contrast

While the mean tells us about brightness, the spread of the histogram tells us about **contrast**. This is mathematically quantified by the statistical moments about the mean. The **Variance** (σ^2) is the second central moment:

$$\sigma^2 = \sum_{k=0}^{L-1} (r_k - \mu)^2 p(r_k)$$

The **Standard Deviation** ($\sigma = \sqrt{\sigma^2}$) describes the average deviation of pixels from the mean brightness.

- **Low σ** : The histogram is narrow. Pixels are similar in value. The image looks flat or washed out (Low Contrast).
- **High σ** : The histogram is wide. There is a significant difference between dark and light areas (High Contrast).

2.4 The Cumulative Distribution Function (CDF)

The Cumulative Distribution Function (CDF) is crucial for advanced techniques like Histogram Equalization. While the PDF $p(r_k)$ tells us the probability of a specific value, the CDF $P(r_k)$ tells us the probability of observing a value *less than or equal to* r_k .

$$P(r_k) = \sum_{j=0}^k p(r_j)$$

The CDF is a monotonically non-decreasing function where $P(r_0) = p(r_0)$ and $P(r_{L-1}) = 1$. In image processing, the CDF essentially maps the rank of a pixel intensity relative to the population.

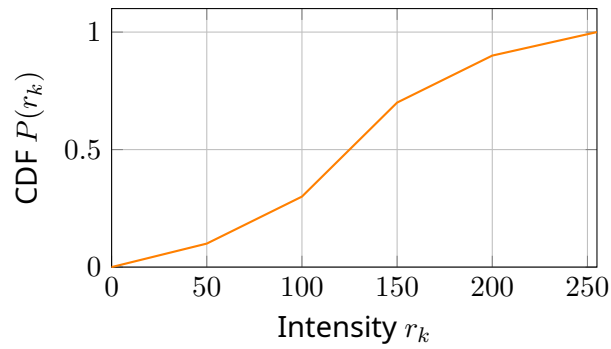


Figure 2.2: A typical CDF curve for a general image.

2.5 Pixel Normalization: Min-Max Scaling

In machine learning and neural networks, raw pixel values in $[0, 255]$ can cause numerical instability because they are large integers. Min-Max normalization linearly transforms the data to a bounded range, typically $[0, 1]$.

$$s = \frac{r - r_{min}}{r_{max} - r_{min}}$$

If we assume the theoretical range ($r_{min} = 0, r_{max} = 255$), this simplifies to just dividing by 255. This interpretation treats pixel intensity as a "presence percentage" of light.

2.6 Standardization: Z-Score Normalization

Unlike Min-Max scaling, which bounds the data, Standardization centers the data distribution. It transforms the image such that the new mean is 0 and the new standard deviation is 1.

$$s = \frac{r - \mu}{\sigma}$$

This is particularly important for Kernel-based classifiers (like SVMs) or Convolutional Neural Networks, which assume that input features are centered around zero to ensure efficient gradient descent convergence. A standardized image will have approximately half its pixels negative (darker than average) and half positive (brighter than average).

2.7 Image Entropy: Quantifying Information

Entropy is a concept borrowed from Information Theory. It measures the amount of information or "randomness" in an image. An image with uniform color (e.g., a pure blue sky) has very low entropy because it is predictable. An image with complex texture has high entropy.

$$H = - \sum_{k=0}^{L-1} p(r_k) \log_2 p(r_k)$$

Entropy is maximized when the histogram is perfectly uniform (flat). This is the theoretical basis for **Histogram Equalization**: we modify the image to maximize its information content (entropy) by flattening its histogram.

2.8 Multivariate Statistics: Color Histograms

For RGB images, simple 1D histograms can be misleading. A pixel is a vector $\mathbf{v} = [R, G, B]^T$. We can analyze:

- **Channel Histograms:** analyzing R , G , and B distributions independently. This is useful for color balancing (e.g., detecting if an image is too "red").
- **Joint Histograms:** A 3D histogram where bins represent combinations of (r, g, b) . This captures correlations, such as the fact that in a landscape, High Blue often correlates with High White (sky/clouds).

Chapter 3

The Calculus of Pixel Transformation

Image enhancement is the art and science of improving the quality of an image for human perception or machine analysis. Mathematically, these techniques are point processing operations, meaning the new value s of a pixel depends solely on its current value r . We denote this as $s = T(r)$. We explore the six fundamental families of enhancement transformations.

3.1 Linear Transformations: Brightness & Contrast

The simplest form of enhancement involves linear manipulation of pixel intensities. This is governed by the affine equation:

$$s = \alpha r + \beta$$

Here, α represents the gain (contrast control) and β represents the bias (brightness control).

- **Brightness (β):** Adding a constant β shifts the entire histogram. If $\beta > 0$, the image brightens; if $\beta < 0$, it darkens.
- **Contrast (α):** Multiplying by α stretches or shrinks the dynamic range. If $\alpha > 1$, intensity differences increase (higher contrast).

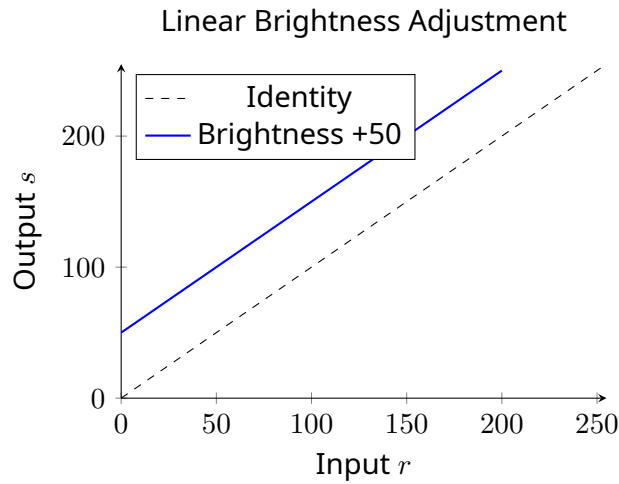


Figure 3.1: Adding a bias term shifts the transfer function upward.

3.2 Image Inversion (Negatives)

The negative of an image is the digital equivalent of a photographic negative. For an image with intensity levels L (typically 256), the transformation is:

$$s = L - 1 - r$$

This operation reverses the pixel values: black (0) becomes white (255) and white becomes black. This technique is particularly vital in medical imaging, such as mammography, where small details (micro-calcifications) are easier to analyze against a dark background than a bright one.

3.3 Thresholding (Binarization)

Thresholding is an extreme form of contrast enhancement that converts a grayscale image into a binary image (black and white). It is the foundation of image segmentation. The transformation function is a step function defined by a threshold value T :

$$s = \begin{cases} 255 & \text{if } r > T \\ 0 & \text{if } r \leq T \end{cases}$$

This operation isolates objects of interest from the background. Mathematically, it can be seen as applying the Heaviside step function.

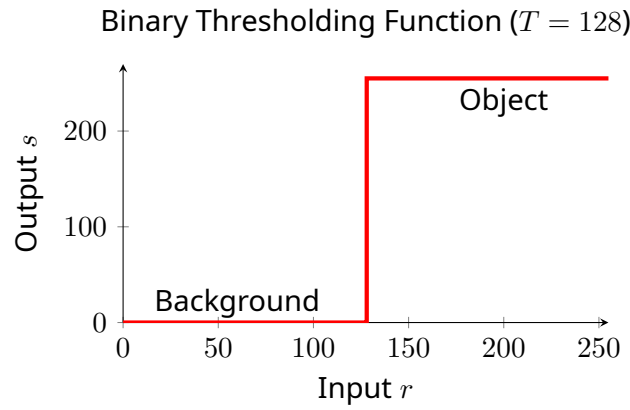


Figure 3.2: The step function classifies pixels into two sets: object or background.

3.4 Contrast Stretching

Low-contrast images result from poor illumination or sensor sensitivity. Contrast stretching expands the range of intensity levels to span the full dynamic range $[0, 255]$. Let r_{min} and r_{max} be the lowest and highest pixel values in the original image. The transformation is:

$$s = 255 \cdot \frac{r - r_{min}}{r_{max} - r_{min}}$$

This linear normalization ensures that the darkest pixel becomes pure black (0) and the brightest becomes pure white (255), maximizing visual dynamic range.

3.5 Bit-Plane Slicing

An 8-bit image is composed of 8 bit-planes, ranging from bit 0 (Least Significant Bit, LSB) to bit 7 (Most Significant Bit, MSB). The intensity of a pixel is given by the polynomial:

$$r = \sum_{n=0}^7 k_n \cdot 2^n$$

Bit-plane slicing isolates specific bits to analyze their contribution. The MSB (bit 7) contains the majority of the structural information, while the LSB (bit 0) often contains random noise. By reconstructing an image using only high-order bits (e.g., planes 5, 6, and 7), we can compress the image with minimal perceptual loss.

3.6 Non-Linear Transformations

Linear scaling cannot recover visibility in shadows without washing out highlights. We require non-linear functions to stretch dynamic range selectively.

3.6.1 Logarithmic Transformation

The logarithmic transformation is given by:

$$s = c \log(1 + r)$$

where c is a scaling constant $c = \frac{255}{\log(1+\max(r))}$.

This transformation maps a narrow range of low-intensity values in the input to a wider range of output levels. It is extensively used in displaying Fourier spectra, where values can range from 0 to 10^6 . Without this compression, the spectrum would appear almost entirely black.

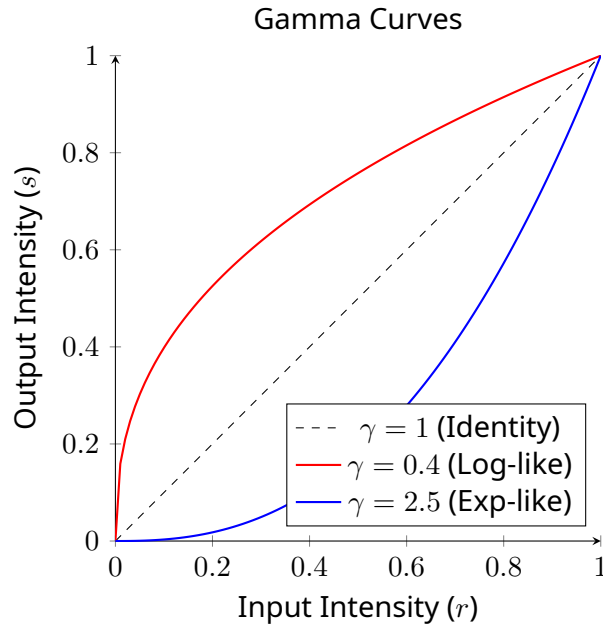
3.6.2 Power-Law (Gamma) Transformation

Gamma correction is a versatile technique governed by the power-law equation:

$$s = cr^\gamma$$

This function offers a flexible family of curves:

- $\gamma < 1$ (**e.g., 0.4**): Expands dark values. This “brightens” the image and reveals shadow detail. This is commonly used in medical imaging to view bone structures.
- $\gamma > 1$ (**e.g., 2.5**): Compresses dark values and expands bright values. This “darkens” the image and is useful for washing out gray background noise.



3.6.3 Sigmoidal Transformation (Contrast S-Curve)

Often used in deep learning data augmentation, the sigmoid function gently compresses extremes (shadows and highlights) while expanding the mid-tones.

$$s = \frac{1}{1 + e^{-\alpha(r-\beta)}}$$

By centering the sigmoid (β) at the mean intensity of the image, we can significantly boost contrast in the mid-range where most visual information resides, unlike thresholding which destroys all gradient information.

3.7 Image Enhancement Case Study: Histogram Equalization

Histogram equalization is a sophisticated technique that automatically determines a transformation function to produce a uniform histogram. This maximizes the global contrast.

Mathematical Logic: We seek a transformation $s = T(r)$ such that the output Probability Mass Function is uniform: $p_s(s) = \frac{1}{L-1}$. The solution relies on the Cumulative Distribution Function (CDF).

Problem Data: Consider a 3-bit image (values 0-7) with dimensions 4×4 ($N = 16$). The intensity counts are:

Intensity (r_k)	0	1	2	3	4	5	6	7
Count (n_k)	0	0	0	6	8	2	0	0

Notice the low contrast; pixels are clustered at 3, 4, and 5.

Step 1: Calculate PMF $p(r_k)$ and CDF. The Cumulative Distribution Function (CDF) is defined as $CDF(r_k) = \sum_{j=0}^k p(r_j)$.

$$p(3) = 6/16 = 0.375, \quad p(4) = 8/16 = 0.5, \quad p(5) = 2/16 = 0.125$$

Cumulative probabilities:

- $CDF(0) \dots CDF(2) = 0$
- $CDF(3) = 0.375$
- $CDF(4) = 0.375 + 0.5 = 0.875$
- $CDF(5) = 0.875 + 0.125 = 1.0$

Step 2: Map to new range. The formula for the equalized value s_k is $s_k = \text{round}((L-1) \times CDF(r_k))$, where $L = 8$ (max intensity is 7).

- For $r = 3$: $s = \text{round}(7 \times 0.375) = \text{round}(2.625) = 3$
- For $r = 4$: $s = \text{round}(7 \times 0.875) = \text{round}(6.125) = 6$
- For $r = 5$: $s = \text{round}(7 \times 1.0) = 7$

Result: The pixels originally at intensity 3 stay at 3. Pixels at 4 jump to 6. Pixels at 5 jump to 7. The dynamic range has expanded from $[3, 5]$ to $[3, 7]$, enhancing the contrast significantly.

Chapter 4

Linear Algebra of Convolution & Classification

4.1 The Mathematical Engine: Convolution

Convolution is a sliding dot product. It detects features by matching a kernel pattern K against image patches.

$$S(x, y) = (I * K)(x, y) = \sum_i \sum_j K(i, j) \cdot I(x - i, y - j)$$

4.2 The Menagerie of Kernels

Different kernels (K) extract different features.

- **Identity:** A center 1 with surrounding 0s. Leaves image unchanged.
- **Box Blur:** All values are $1/9$. Averages neighbors, smoothing noise.
- **Edge Detection (Sobel):** Approximates gradients.

$$K_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

4.3 Dimensionality Reduction: Pooling

Convolutional features are often high-dimensional. Pooling layers reduce the spatial size of the representation to reduce the amount of parameters and computation in the network.

4.3.1 Max Pooling

Max pooling is the most common. It partitions the input image into a set of non-overlapping rectangles and outputs the maximum value for each sub-region.

$$y_{i,j} = \max_{(p,q) \in \mathcal{R}_{i,j}} x_{p,q}$$

This provides translation invariance: even if a feature (like an edge) shifts slightly, the max value in the window remains the same.

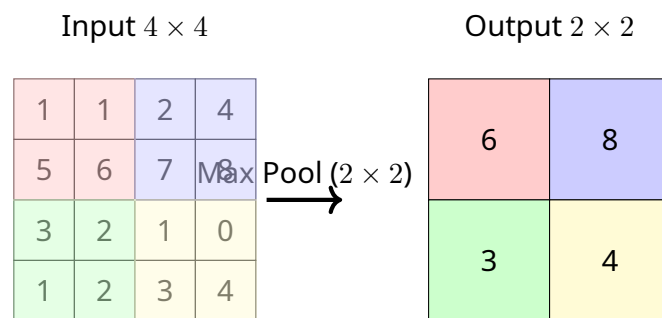


Figure 4.1: Visualizing Max Pooling. The grid is divided into 2×2 blocks, and only the largest number survives.

4.3.2 Average Pooling

Average pooling calculates the mean of each patch. It effectively smooths the image, similar to a blur filter, but reduces resolution. It preserves background information better than max pooling but can lower contrast.

4.4 Kernel-Based Classification Study Case

Problem: Classify whether a 5×5 image is a "Horizontal Line" or "Vertical Line".

Linear Algebra Pipeline:

1. **Input vector $\mathbf{x}$:** Flatten the 5×5 image into $\mathbb{R}^{25}$.
2. **Weights $\mathbf{w}$:** The flattened kernel acts as a weight vector.
3. **Score z :** $z = \mathbf{w}^T \mathbf{x} + b$.
4. **Activation:** $y = \text{sigmoid}(z) = \frac{1}{1+e^{-z}}$.

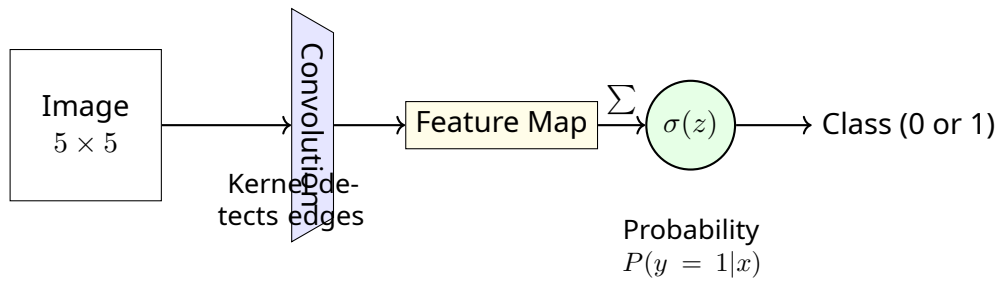


Figure 4.2: The Classical Classification Pipeline

Chapter 5

Image Augmentation

Data augmentation artificially expands the training dataset by applying semantics-preserving transformations.

5.1 Affine Transformations

Affine transformations preserve collinearity and ratios of distances.

5.1.1 Translation

Translation moves the image by a vector $\mathbf{t} = [t_x, t_y]^T$.

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

The "empty" space left behind is usually filled with black (0) or reflection padding.

5.1.2 Scaling (Zooming)

Scaling resizes the image.

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

5.1.3 Rotation

To rotate an image by an angle θ around the origin:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

5.2 Cropping

Random cropping extracts a sub-region $I[y : y + h, x : x + w]$. This forces the model to focus on specific features rather than the object's absolute position.

5.3 Morphological Augmentations

While geometric transformations move pixels, morphological operations change shapes based on structure.

5.3.1 Dilation

Dilation expands bright regions (features). Mathematically, it is the vector addition of set elements.

$$(A \oplus B)(z) = \max\{A(z - x) + B(x)\}$$

It fills in holes and connects disjoint features.

5.3.2 Erosion

Erosion shrinks bright regions (expands dark regions). It is useful for removing small salt noise or detaching bridged objects.

$$(A \ominus B)(z) = \min\{A(z + x) - B(x)\}$$

5.4 Conclusion

We have visually and mathematically traversed the pipeline of image processing. From the grid representation to 3D color spaces, histogram probability distributions, non-linear gamma curves, and finally, the sliding window mechanism of convolution. These visualizations serve to ground the abstract algebra in concrete geometric intuition.